

Manual and Automated Mutation Testing

Problem Statement

High code coverage does not necessarily indicate an effective test suite. Mutation testing evaluates the quality of test cases by introducing small changes (mutants) into the source code and measuring whether the existing tests detect these changes.

Develop a mutation testing workflow that performs both **manual** and **automated mutation testing**, evaluates the effectiveness of the test suite, identifies surviving mutants, and generates a mutation testing report with an overall mutation score.

** The required code is provided in resources.*

Objective

The objective of this assignment is to:

- Understand the concepts of mutation testing.
- Evaluate the effectiveness of unit test suites.
- Create and analyze software mutants.
- Build an automated mutation testing pipeline.
- Improve test quality by increasing the mutation score.

Features to Implement

- Develop unit tests for the given software.
- Create at least twenty manual mutants.
- Execute tests against each mutant.
- Identify killed and surviving mutants.
- Improve the test suite based on mutation analysis.

- Achieve a mutation score greater than 95%.
- Build an automated mutation generation pipeline.
- Execute tests automatically for generated mutants.
- Calculate and report the overall mutation score.
- Generate mutation testing reports and analysis.